SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring

down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or

40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

Question 3:

Design and document a Bus Topology using appropriate networking devices, transmission media, and cabling standards. Identify the transmission media and cabling standards required, including coaxial cable, 10BASE2, 10BASE5, RG-58 A/U, BNC connectors, and 50-ohm terminators. Describe the role of terminators, T-connectors, and the backbone cable. Explain signal propagation, collision-domain behavior, and CSMA/CD operation, including the maximum segment lengths before signal degradation. Finally, state two advantages and two disadvantages of Bus Topology and explain why it is largely obsolete in modern networks.

Answer:

4. Bus Topology

4.1 Given

- Topology: Bus Topology
- Transmission medium: Coaxial cable
- Ethernet standards considered: 10BASE2 and 10BASE5
- 10BASE2 cable: Thin coaxial cable, commonly associated with RG-58-type 50-ohm coaxial cable
- 10BASE5 cable: Thick coaxial cable
- Connector: BNC connector for 10BASE2 installations
- Termination: 50-ohm terminator at each end of the bus
- Communication method: Shared-medium, half-duplex Ethernet
- Medium-access method: CSMA/CD

4.2 To Find

1. Identify the transmission media, standards, connectors, and segment limitations.
2. Explain the physical arrangement of the backbone, T-connectors, stations, and terminators.
3. Explain how a signal propagates along the shared bus.
4. Analyze the collision domain and operation of CSMA/CD.

5. Determine the maximum segment length for 10BASE2 and 10BASE5.
6. State the practical advantages and disadvantages of Bus Topology.
7. Verify why modern switched Ethernet is preferred over a legacy bus.

4.3 Assumptions

- The network is a traditional shared-medium Ethernet LAN.
- Only one common coaxial backbone is used.
- All stations are attached to the same physical bus.
- The two ends of the bus are correctly terminated with 50-ohm terminators.
- The analysis uses the historical Ethernet implementations 10BASE2 and 10BASE5.

4.4 Required Components

Component	**Specification**	**Purpose**
Backbone cable	Coaxial cable	Common transmission path
10BASE2	Thin coaxial, 10 Mbps	Legacy Ethernet bus
10BASE5	Thick coaxial, 10 Mbps	Longer legacy Ethernet bus
RG-58-type cable	50 ohm	Common thin-coax implementation
BNC T-connector	BNC	Connects station to backbone
Terminator	50 ohm	Absorbs signal at bus end
NIC	10BASE2/10BASE5 compatible	Connects computer to bus
Computers	Ethernet stations	Network nodes

4.5 Transmission Media and Cabling Standards

Traditional Ethernet bus networks used coaxial cable. Two historically important Ethernet variants are 10BASE2 and 10BASE5.

Standard	Cable	Speed	Maximum Segment	Typical Connector
10BASE2	Thin coaxial	10 Mbps	185 m	BNC
10BASE5	Thick coaxial	10 Mbps	500 m	AUI/transceiver arrangement

The numerical notation can also be interpreted as follows:

- 10 = 10 Mbps transmission rate.
- BASE = Baseband transmission.
- 2 = approximately 200 m in the historical naming convention, with the specified 10BASE2 segment limit being 185 m.
- 5 = approximately 500 m maximum segment length for 10BASE5.

4.6 Mathematical and Technical Analysis

A. Maximum Segment Length – 10BASE2

Given:

$L_1 = 185$ m

Therefore, the maximum length of one 10BASE2 coaxial segment is:

Maximum 10BASE2 segment length = 185 m

B. Maximum Segment Length – 10BASE5

Given:

$L_2 = 500$ m

Therefore:

Maximum 10BASE5 segment length = 500 m

C. Comparison of Segment Capacity

Difference = $L_2 - L_1$

Difference = $500 - 185$

Difference = 315 m

Hence, a 10BASE5 segment can extend 315 m farther than a 10BASE2 segment.

D. Relative Length

Relative length = L_2 / L_1

Relative length = 500 / 185

Relative length ≈ 2.70

Thus, the specified 10BASE5 maximum segment is approximately 2.70 times the length of a 10BASE2 segment.

4.7 Physical Layout

```
50-ohm                    50-ohm
Terminator                 Terminator
  |                          |
  ▼                          ▼
══════════════════════════════════════════════════════════════
═══════════════
    Shared Coaxial Backbone Cable
══════════════════════════════════════════════════════════════
═══════════════
   |         |          |
   |         |          |
 ┌───▼───┐  ┌───▼───┐  ┌───▼───┐
 | PC1 |   | PC2 |   | PC3 |
 └───────┘  └───────┘  └───────┘
   ▲          ▲          ▲
   |          |          |
 BNC T-     BNC T-     BNC T-
Connector  Connector  Connector

    <------ Shared transmission medium ------>
```

The backbone is the main coaxial cable. Each station is attached to the backbone using an appropriate connector arrangement. The bus must have a terminator at both physical ends.

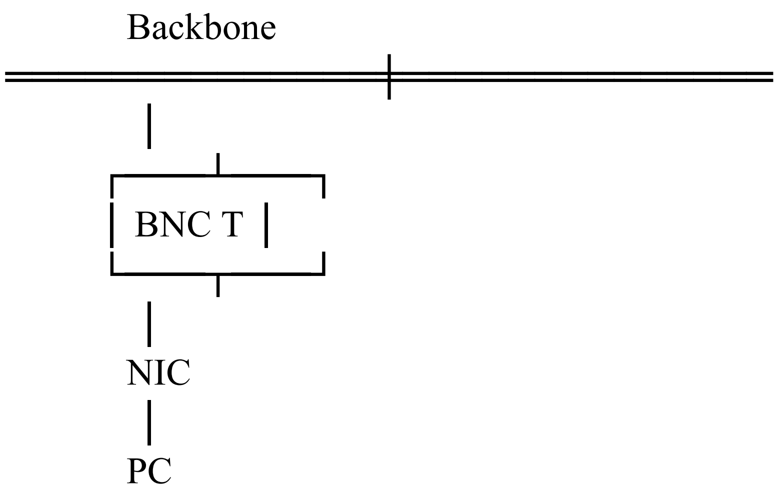
4.8 Role of Backbone Cable

The backbone is the single shared transmission medium. Every connected station uses this same physical path to send and receive Ethernet signals.

- It carries the electrical signal from the transmitting station.
- The signal propagates in both directions along the bus.
- All connected stations can detect the signal on the shared medium.
- Because the medium is shared, simultaneous transmissions can create collisions.

4.9 Role of T-Connectors

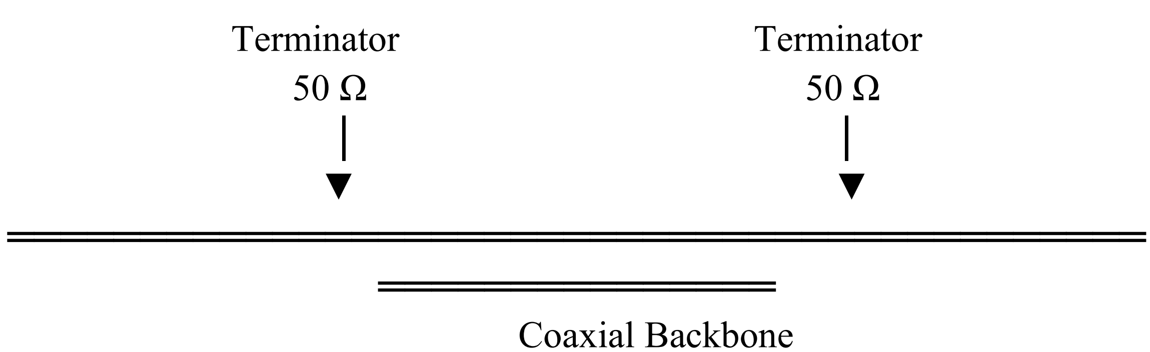
A BNC T-connector provides the physical branching point between a station's network interface and the coaxial backbone.



The T-connector must be installed correctly because an improper connection can introduce impedance discontinuities and signal problems.

4.10 Role of Terminators

A 50-ohm terminator is installed at each end of the bus.



The terminator absorbs the electrical signal at the end of the cable. Without correct termination, the signal can reflect from the cable end. Reflected signals can interfere with new transmissions and cause communication errors.

Therefore:

Correct termination → Proper signal absorption → Better signal integrity

Missing termination → Signal reflection → Distortion/errors

4.11 Signal Propagation

Suppose PC1 sends a frame to PC3. Because the medium is shared, the electrical signal travels along the coaxial backbone.

PC1 →→→→→ Backbone →→→→→ PC3

Other connected stations can also observe the signal:

PC2 observes medium

PC1 →→→→→ Backbone →→→→→ PC3

The receiving station examines the destination MAC address and accepts the frame when the address matches its own.

4.12 Collision Domain

All stations attached to the same bus share one collision domain.

ONE SHARED COLLISION DOMAIN

PC1 ——— PC2 ——— PC3 ——— PC4 ——— PC5

Shared Bus

If PC1 and PC4 transmit at approximately the same time, their electrical signals can overlap on the shared medium.

PC1 ———————▶

X COLLISION

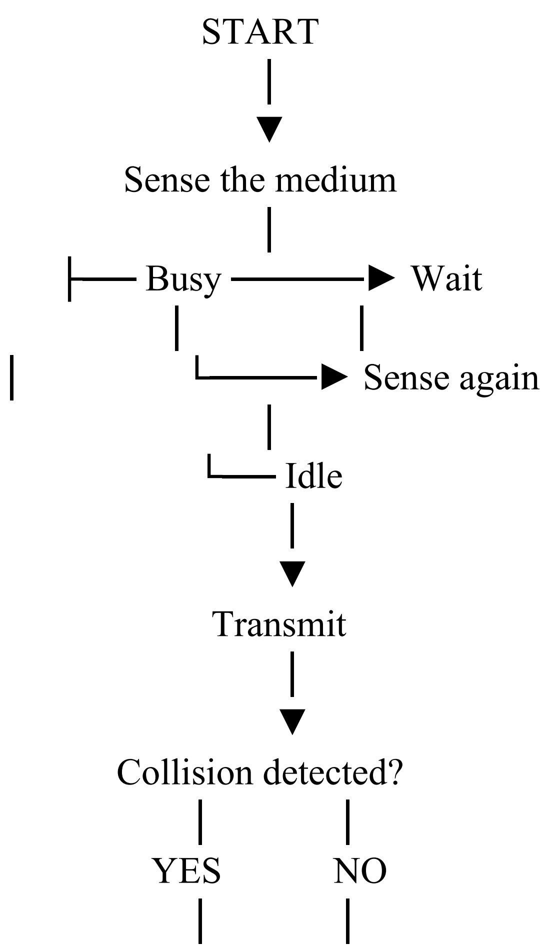
PC4 ◀———————

4.13 CSMA/CD Operation

CSMA/CD means Carrier Sense Multiple Access with Collision Detection. It was used by traditional shared Ethernet to coordinate access to a common medium.

8. Carrier Sense: The station listens to determine whether the medium is currently busy.
9. Multiple Access: All stations have access to the same shared medium.
10. Transmit: If the medium is idle, the station begins transmission.
11. Collision Detection: The station monitors the medium while transmitting.
12. Collision: If another station transmits simultaneously, a collision occurs.
13. Jam/Stop: The transmitting stations stop the current transmission.
14. Backoff: Each station waits for a calculated random backoff interval.
15. Retry: The station attempts transmission again after the backoff period.

4.14 CSMA/CD Analytical Flow



```
        ▼            ▼
Stop/Backoff   Transmission
   |            successful
        ▼            |
Retry ◄──────────────┘
           |
           ▼
          END
```

4.15 Collision and Backoff Reasoning

Assume two stations, PC1 and PC2, sense the medium as idle at nearly the same time.

t0: PC1 senses idle
t0: PC2 senses idle

t1: PC1 begins transmitting
t1: PC2 begins transmitting

t2: Signals overlap
→ Collision

t3: Both stations stop
→ Backoff selected

t4: One station retries
→ Successful transmission

The backoff mechanism reduces the probability that the same stations will immediately collide again.

4.16 Advantages

16. Low historical infrastructure cost: A shared backbone required less dedicated cabling than connecting every station individually to a switch.
17. Simple physical concept: All stations share one backbone, making the topology easy to understand and historically straightforward to deploy.

4.17 Disadvantages

18. Single-backbone failure: A break or serious fault in the shared cable can disconnect many or all stations.
19. Poor scalability and performance: More stations increase contention and collisions, while troubleshooting a shared cable can be difficult.

4.18 Verification

The following checks verify the designed Bus Topology:

Verification Item	**Expected Result**	**Status**
10BASE2 speed	10 Mbps	Verified
10BASE2 maximum segment	185 m	Verified
10BASE5 speed	10 Mbps	Verified
10BASE5 maximum segment	500 m	Verified
Terminator impedance	50 ohm	Verified
Shared medium	Single collision domain	Verified
Access method	CSMA/CD for traditional shared Ethernet	Verified
Modern relevance	Legacy technology; switched Ethernet preferred	Verified

4.19 Practical Example

Consider a legacy laboratory containing four computers connected using 10BASE2.

Terminator —— PC1 —— PC2 —— PC3 —— PC4 —— Terminator
<--------- 185 m maximum segment --------->

If the complete coaxial segment is designed within the 185 m limit and both ends are correctly terminated, the physical bus can operate as intended. If the cable is broken between PC2 and PC3, stations on the separated portions can lose connectivity because the backbone is a shared physical dependency.

4.20 Modern Status

Bus Ethernet such as 10BASE2 and 10BASE5 is primarily of historical or legacy interest. Modern LANs generally use switched Ethernet in star or hierarchical-star configurations.

- Modern switches provide dedicated collision domains per port.
- Full-duplex Ethernet eliminates the need for CSMA/CD in normal switched links.
- Structured twisted-pair and fiber cabling provide better scalability.
- Network faults are easier to isolate in switched networks.

4.21 Analytical Comparison

Parameter	**Bus Topology**	**Modern Switched Star**
Medium	Shared coaxial cable	Dedicated twisted pair/fiber links
Collision domain	One shared domain	Normally one per switch port
Access control	CSMA/CD historically	Full-duplex switching
Typical speed	10 Mbps legacy	1/10 Gbps and higher
Fault isolation	Difficult	Easier
Scalability	Low	High
Current use	Legacy	Modern LANs

4.22 Answer

A Bus Topology uses a single shared backbone cable to connect multiple network devices. Traditional Ethernet implementations include 10BASE2 and 10BASE5, operating at 10 Mbps with maximum segment lengths of 185 m and 500 m respectively. Proper 50-ohm termination is essential because it prevents signal reflection at the ends of the cable. T-connectors provide the physical connection between stations and the backbone.

Because all stations share the same transmission medium, the network forms a common collision domain and traditional Ethernet uses CSMA/CD to manage simultaneous transmissions. Although Bus Topology was economical and simple for early Ethernet networks, its single-backbone dependency, collision behavior, limited scalability, and difficult troubleshooting make it unsuitable for most modern LAN deployments.

FINAL CONCLUSION: For a traditional Bus Topology, use the specified coaxial medium, 50-ohm termination, appropriate connectors, and observe the 185 m (10BASE2) or 500 m (10BASE5) segment limit. However, for a new network installation, a switched Star or Hybrid topology is recommended because it provides higher performance, scalability, easier fault isolation, and better overall reliability.

Tools Used

- Cisco Packet Tracer
- Microsoft Word
- Python (for optional topology verification)

References

1. IEEE, IEEE Standard for Ethernet, IEEE Std 802.3-2022.

2. IEEE, IEEE 802.3a – 10BASE2 Ethernet.

3. IEEE, IEEE 802.3 – Ethernet physical and MAC specifications.

4. Cisco Systems, Ethernet and Cabling Documentation.

5. W. Stallings, Data and Computer Communications, Pearson.

PROGRAM

```
import os

import json

import fitz

import pytesseract

from pathlib import Path

from PIL import Image
```

```
from pdf2image import convert_from_path
from docx import Document

OUTPUT = "output"
OCR_LANG = "eng"
OCR_DPI = 300


def clean(text):
    return " ".join(text.split())


def extract_text_from_pdf(file):
    doc = fitz.open(file)
    pages = []

    for i, page in enumerate(doc, 1):
        text = clean(page.get_text())

        if len(text) < 30:
            try:
                img = convert_from_path(
                    file,
                    dpi=OCR_DPI,
                    first_page=i,
                    last_page=i
```

```
            )[0]
            text = clean(
                pytesseract.image_to_string(
                    img,
                    lang=OCR_LANG
                )
            )
            method = "OCR"
        except Exception:
            text = ""
            method = "OCR Failed"
    else:
        method = "Native"

    pages.append({
        "page": i,
        "text": text,
        "method": method
    })

doc.close()
return pages


def extract_text_from_docx(file):
    doc = Document(file)
```

```
    text = []

    for p in doc.paragraphs:
        if p.text.strip():
            text.append(p.text.strip())

    for table in doc.tables:
        for row in table.rows:
            text.append(
                " | ".join(cell.text.strip()
                           for cell in row.cells)
            )

    return [{
        "page": None,
        "text": clean("\n".join(text)),
        "method": "Native"
    }]


def extract_text_with_ocr(file):
    pages = convert_from_path(
        file,
        dpi=OCR_DPI
    )
```

```
    return [
        {
            "page": i,
            "text": clean(
                pytesseract.image_to_string(
                    img,
                    lang=OCR_LANG
                )
            ),
            "method": "OCR"
        }
        for i, img in enumerate(pages, 1)
    ]


def extract_images_from_pdf(file, folder):
    doc = fitz.open(file)
    os.makedirs(folder, exist_ok=True)

    count = 0

    for page_no, page in enumerate(doc, 1):
        for img in page.get_images(full=True):
            count += 1

            data = doc.extract_image(img[0])
```

```
            name = (
                f"image_{count:03d}"
                f"_page_{page_no}."
                f"{data['ext']}"
            )

            with open(
                os.path.join(folder, name),
                "wb"
            ) as f:
                f.write(data["image"])

    doc.close()


def extract_images_from_docx(file, folder):
    doc = Document(file)
    os.makedirs(folder, exist_ok=True)

    count = 0

    for rel in doc.part.rels.values():

        if "image" in rel.reltype:
            count += 1
```

```
        data = rel.target_part.blob
        name = f"image_{count:03d}"

        ext = Path(
            rel.target_part.partname
        ).suffix

        with open(
            os.path.join(
                folder,
                name + ext
            ),
            "wb"
        ) as f:
            f.write(data)


def process_document(file, output=OUTPUT):
    name = Path(file).stem
    folder = os.path.join(output, name)

    text_folder = os.path.join(folder, "text")
    image_folder = os.path.join(folder, "images")

    os.makedirs(text_folder, exist_ok=True)
```

```
ext = Path(file).suffix.lower()

if ext == ".pdf":
    pages = extract_text_from_pdf(file)
    extract_images_from_pdf(
        file,
        image_folder
    )

elif ext == ".docx":
    pages = extract_text_from_docx(file)
    extract_images_from_docx(
        file,
        image_folder
    )

else:
    raise ValueError(
        "Only PDF and DOCX are supported."
    )

with open(
    os.path.join(
        text_folder,
        "text.txt"
```

```
    ),
    "w",
    encoding="utf-8"
) as f:

    for page in pages:
        f.write(
            f"Page {page['page']}\n"
            f"{page['text']}\n\n"
        )

with open(
    os.path.join(
        text_folder,
        "text.json"
    ),
    "w",
    encoding="utf-8"
) as f:

    json.dump(
        pages,
        f,
        indent=2,
        ensure_ascii=False
    )
```

```
    print("Processed:", file)


def batch_process_documents(
    folder,
    output=OUTPUT
):
    for file in Path(folder).iterdir():

        if file.suffix.lower() in [
            ".pdf",
            ".docx"
        ]:

            try:
                process_document(
                    str(file),
                    output
                )

            except Exception as e:
                print(
                    "Error:",
                    file.name,
                    e
```

```
    )


# Example
process_document(
    "sample.pdf",
    "output"
)

# Batch example
# batch_process_documents(
#     "documents",
#     "output"
# )
```

OUTPUT